Window-Adaptive, Multi-Objective Metaheuristic Co-Optimization of Features and Hyperparameters for Partial-Charging Battery SOH Regression

1 Introduction

State of Health (SOH) estimation is central to battery management in electric vehicles, where it supports safety, range prediction and maintenance planning. Data-driven models estimate SOH from features extracted from voltage, current and temperature measurements. Most laboratory studies rely on full 0–100% charge cycles, whereas real vehicles typically charge within partial State-of-Charge ranges. Partial-charging features such as Incremental Capacity (dQ/dV) curves, differential temperature and voltage entropy are therefore a more realistic basis for SOH estimation.

Two difficulties remain. First, many candidate features are correlated and redundant, so feature selection and model hyperparameter tuning must be handled together rather than separately. Second, laboratory datasets contain only a few cells, so models that look accurate on one split often fail on unseen cells. This project builds a leakage-safe optimization framework for SOH regression from partial-charging data and tests rigorously whether joint optimization improves generalization to unseen cells.

2 Novelty

Standard battery SOH models assume full-cycle data and treat feature selection and hyperparameter tuning as separate steps. Recent work (Makama & Thango, Oct 2026) couples metaheuristic feature selection with tree-ensemble tuning on field vehicle data, but it reports that a fixed default ensemble outperformed the optimized hybrids on held-out capacity, so the benefit of joint optimization is unresolved. Existing pipelines also fix the partial-charging window by hand rather than learning it, and they rarely optimize for robustness across cells (Su et al., Feb 2026). The novelty of this project is threefold:

1. The partial-charging voltage window itself becomes a searchable variable, optimized jointly with the feature mask and regressor hyperparameters.
2. A Pareto-based multi-objective search replaces weighted-sum scalarization, with worst-cell error and cross-cell error variance as explicit objectives alongside feature fraction and model complexity.
3. The framework is tested as a hypothesis: does joint co-optimization beat equally tuned simple baselines under leave-one-cell-out evaluation and cross-dataset transfer (NASA $\rightarrow$ Oxford)?

3 Methodology

1. **Window-Adaptive Partial-Cycle Feature Expansion:** Extract 40+ candidate health indicators (dQ/dV peak coordinates/areas, dT/dV, voltage sample entropy) from a constant-current charging segment bounded by $[V_{\text{start}}, V_{\text{end}}]$. Both bounds are encoded as continuous

genes in the search vector, constrained to a minimum window width and to realistic partial-charging ranges (e.g., within 3.6 V–4.1 V).

2. **Leakage-Safe Leave-One-Cell-Out Evaluation:** Because both datasets contain few cells, evaluate with leave-one-cell-out cross-validation instead of a single split. The metaheuristic search runs only on training cells, with an inner cell-disjoint validation fold. The held-out cell is used once, after the final configuration is selected, to prevent validation-set overfitting by the optimizer.

3. **Pareto Multi-Objective Co-Optimization:** Encode the D-dimensional feature mask, the window bounds and the regressor hyperparameters in one search vector. Optimize with an NSGA-II-style Pareto search and a hybrid metaheuristic (Adaptive DE–PSO or HHO–GWO) over four objectives:

   (a) mean cross-cell RMSE,
   (b) worst-cell RMSE (or cross-cell error variance),
   (c) feature fraction $N_{\text{selected}}/N_{\text{total}}$,
   (d) model complexity.

   The final solution is chosen from the Pareto front by a knee-point rule.

4. **Regression Model:** Predict continuous SOH (%) with a stacked ensemble (Support Vector Regression + Random Forest / XGBoost with a Ridge meta-learner). To keep the search tractable, only the key hyperparameters of each base learner are searched, and the stacking structure is fixed. Remaining Useful Life (RUL) prediction is a secondary extension.

5. **Fair Baselines and Statistical Testing:** Benchmark against LASSO (L_1), SHAP-based Recursive Feature Elimination and Mutual Information, each paired with a tuned regressor. Add random search and TPE (e.g., Optuna) at the same evaluation budget as the metaheuristics, plus a fixed default-hyperparameter model. Report mean ± standard deviation over repeated seeds, paired Wilcoxon tests with Holm correction, and negative results.

6. **Cross-Dataset Transfer and Uncertainty:** Train on NASA cells and test on Oxford cells (and vice versa) to evaluate transferability across operating profiles. Optionally, add conformal prediction intervals to give SOH estimates calibrated uncertainty bounds.

4 Datasets

- **Data Type:** Multivariate time-series electrochemical cycling data (continuous regression for capacity fade and SOH %).
- **NASA Battery Aging Dataset:** Lithium-ion cells cycled to end of life, with voltage, current, temperature and capacity records per cycle. A small number of cells is available, which motivates leave-one-cell-out evaluation.
  - Link: `https://data.nasa.gov/dataset/li-ion-battery-aging-datasets`
  - Cleaned Kaggle mirror: `https://www.kaggle.com/datasets/mystifoe77/nasa-battery-data-cleane`
- **Oxford Battery Degradation Dataset:** Aging data for 8 cells, used as the second dataset for cross-dataset transfer.
  - Link: `https://ora.ox.ac.uk/objects/uuid:03ba4b01-cfed-46d3-9b1a-7d4a7bdf6fac`
- Both datasets must be checked during preprocessing for usable constant-current charging segments, since the window-adaptive features depend on them. Capacity measurements are converted into SOH (%) relative to each cell's initial capacity.